21-242 Recitation 12

Linear Transformations — Q7. Bijectivity iff the image of a basis is a basis

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that $T : V \to W$ is bijective iff $T(\vec{b}_1), \ldots, T(\vec{b}_n)$ is a basis for W.

Course-level solution

T is bijective iff T is injective and surjective iff $\left(T(\vec{b}_i)\right)$ is linearly independent (Q6) and spanning (Q5) iff $\left(T(\vec{b}_i)\right)$ is a basis of W.

Later-course solution (using ideas from the rest of the handout)

If you already know $\dim W = n = \dim V$, any two of independent / spanning / basis for the list $\left(T(\vec{b}_i)\right)$ are equivalent, matching the later fact that injectivity, surjectivity, and bijectivity coincide when $\dim V = \dim W < \infty$.